Slide 1 — Week 5

Database Administration and Management

Topics:

- The DBA role and performance
- Security and access control (Session 9)
- Backup, recovery and capacity (Session 10)
- Data quality, governance and monitoring (Session 10)

Slide 2 — Role of a Database Administrator

Database Administrators (DBAs) manage:

- database performance
- backups and recovery
- security and access control
- database monitoring

Slide 3 — Database Administration Tasks

Common DBA responsibilities:

- install database systems
- configure servers
- manage user permissions
- monitor performance

Slide 4 — Database Performance

Performance tuning involves:

- indexing
- query optimization
- hardware configuration

Slide 5 — Query Optimization

A frequent lookup:

```
SELECT *
FROM orders
WHERE customer_id = 10;
```

A candidate index:

```
CREATE INDEX idx_customer
ON orders(customer_id);
```

Measure before and after — in a columnar engine like DuckDB the index may not help at all, and every index slows writes.

Slide 6 — Database Security

Security protects sensitive data.

Three questions: who may see what, how is sensitive data protected, and how could an attacker get around it?

Common threats:

- unauthorized access
- data leaks
- malicious queries (SQL injection)

Slide 7 — Access Control

Example:

```
CREATE USER 'analyst'@'localhost'
IDENTIFIED BY 'password';
```

Slide 8 — Granting Permissions

```
GRANT SELECT
ON ecommerce.*
TO 'analyst'@'localhost';
```

Slide 9 — Revoking Permissions

```
REVOKE INSERT
ON ecommerce.*
FROM 'analyst'@'localhost';
```

Slide 10 — Access Control in DuckDB: Views

DuckDB is embedded: no user accounts, no GRANT or REVOKE.

Use a **view** as the boundary — expose only some columns and rows:

```
CREATE VIEW v_customer_marketing AS
SELECT customer_id, postal_code, segment   -- no name, no email
FROM customer;
```

⚠️ A view protects data only if the base table is **not** also readable.

Slide 11 — Protecting PII

Treatment	What it does	Joinable?
Redaction	remove the value	no
Tokenization	replace with a stable salted hash	yes
Generalization	reduce precision (age → band)	partly

Removing the name is **not** anonymization — a few quasi-identifiers (postal code, birth date, gender) can re-identify a person.

Slide 12 — SQL Injection

```
# vulnerable
f"SELECT * FROM booking WHERE name = '{user_input}'"

# safe — bind the parameter
"SELECT * FROM booking WHERE name = ?", [user_input]
```

Escaping quotes is not a defence. **Bind every parameter.**

Slide 13 — Backup and Recovery

Backup types:

- full — everything
- incremental — changes since the last backup
- differential — changes since the last full backup
- log shipping — every committed transaction (point-in-time recovery)

Slide 14 — RPO and RTO

RPO — recovery point objective: how much data may we lose?

→ sets backup **frequency**

RTO — recovery time objective: how long may we be down?

→ sets backup **method**

Nightly backups mean losing up to 24 hours of data.

That is a business decision, not just a DBA decision.

Slide 15 — MySQL Backup Example

```
mysqldump -u root -p ecommerce > backup.sql
```

This creates a full database backup.

Slide 16 — Restoring a Database

```
mysql -u root -p ecommerce < backup.sql
```

This restores the database from backup.

Slide 17 — Backup and Restore in DuckDB

```
EXPORT DATABASE 'backup_2026_04_01' (FORMAT PARQUET);
-- ... disaster ...
IMPORT DATABASE 'backup_2026_04_01';
```

The backup is a readable folder: schema.sql, load.sql, one Parquet file per table.

Slide 18 — Test the Restore

A backup you have never restored is a hypothesis.

1. take the backup
2. compute a checksum of the source
3. restore into a clean database
4. re-compute the checksum and compare

Matching row counts is not enough — you can restore the right number of wrong rows.

Slide 19 — Capacity Planning

Measure **bytes**, not rows.

- CSV → Parquet: the value of columnar layout
- uncompressed → zstd: the value of compression

Plan with bytes per row × projected rows, and project the growth rate.

Slide 20 — Data Quality

Six dimensions, each a rule that returns a count:

Dimension	Question
Completeness	is the value present?
Validity	does it match its format?
Uniqueness	one row per real-world thing?
Accuracy	is the value possible?
Timeliness	is it fresh enough?
Consistency	do related facts agree?

Slide 21 — Data Governance

Organizations must manage:

- data quality
- privacy
- compliance

Store governance as tables, so questions become queries:

- **catalog** — every column, with owner, classification, PII flag
- **lineage** — which object feeds which

Slide 22 — Lineage: What Breaks Downstream?

Lineage stored as edges → impact is a **recursive CTE** (the Week 1 pattern):

```
WITH RECURSIVE impact AS (
    SELECT target_object AS obj, 1 AS hops
    FROM lineage WHERE source_object = 'raw_customers'
    UNION ALL
    SELECT l.target_object, i.hops + 1
    FROM lineage AS l JOIN impact AS i ON l.source_object = i.obj
)
SELECT obj, MIN(hops) AS hops FROM impact GROUP BY obj ORDER BY hops;
```

`dim_customer` (1) → `fct_sales` (2) → `rpt_weekly_revenue` (3) → `dash_ceo` (4)

- `MIN(hops)`: an object can be reached by more than one route
- Lineage can contain loops by mistake — stop on revisit

Slide 23 — Master Data: Block, Match, Survive

One real supplier, many records → one **golden record**

1. **Block** — compare only records sharing a cheap key (n^2 is too many)
2. **Match** — `jaro_winkler_similarity()` above a **threshold**; chain A≈B≈C
3. **Survive** — a deterministic survivorship rule picks the winner

a	b	similarity
Northgate Metals	Northgate Metal	0.987
Northgate Metals	Northwind Metals	**0.867** ← different companies!

The threshold trades **wrong merges** against **missed duplicates** — test both.

Slide 24 — Monitoring and the Runbook

A runbook names:

- the metrics and their thresholds
- who is alerted, and when
- the first steps on an alert
- the RPO / RTO you have committed to

Compare metrics against a baseline window, not just yesterday.

Slide 25 — Modern Data Platforms

Organizations run several kinds of database:

- **relational** — MySQL, PostgreSQL, SQL Server: structured data and transactions
- **NoSQL** — MongoDB, Cassandra, DynamoDB: large-scale distributed data
- **cloud** — Amazon RDS, Google BigQuery, Snowflake: managed and scalable

The same administration questions apply to all of them.

Slide 26 — Enterprise Data Architecture

Typical architecture:

Applications

↓

Operational database (OLTP)

↓

Data warehouse

↓

Analytics / dashboards

Slide 27 — Course Summary

During this course you learned:

- data management and conceptual modeling (ER / EER)
- relational design and normalization
- database implementation: DDL, loading, indexes, transactions
- analytical and non-relational data systems
- database administration and management

Slide 28 — What Comes Next: the Final Exam

One exam covering the whole course, in a separately scheduled slot after Session 10.

You will be asked to:

- model a business scenario (ER → relational schema)
- normalize from functional dependencies
- write DDL with the right keys and constraints
- design and query a star schema
- reason about transactions, security, and recovery

Format and rules: see `course_information/ASSIGNMENTS_and_GRADING.md`.